

Ex. 2.34 | Ch. 2 | P. 65 | Chloé | 2025.7.21

解: (a) $E < V_0$ 时反射系数计算

$$x \leq 0 \text{ 时, } V=0, -\frac{\hbar^2}{2m} \frac{d^2\psi_1}{dx^2} = E\psi_1, \psi_1 = Ae^{ik_1x} + Be^{-ik_1x}, k_1 = \sqrt{\frac{2mE}{\hbar^2}}$$

$$x > 0 \text{ 时, } V=V_0, -\frac{\hbar^2}{2m} \frac{d^2\psi_2}{dx^2} + V_0\psi_2 = E\psi_2, \psi_2 = Ce^{-kx} \text{ (舍 } e^{kx}), k = \sqrt{\frac{2m(V_0-E)}{\hbar^2}}$$

又, $x=0$ 时, 波函数连续, ψ 可导, 故 $\psi_1(0) = \psi_2(0) \Rightarrow A+B=C$; $\psi_1'(0) = \psi_2'(0) \Rightarrow$

$$ik_1A - ik_1B = -kC, \text{ 整理得 } B = \frac{ik_1+k}{ik_1-k}A$$

8天出

故 $R = \frac{|B|^2}{|A|^2} = 1$, 这说明 $E < V_0$ 时, 粒子被全部反射, 即使有粒子在势垒中, 但无入(b) $E > V_0$ 时反射系数计算

$$x \leq 0 \text{ 时, } \psi_1 = Ae^{ik_1x} + Be^{-ik_1x}, k_1 = \sqrt{\frac{2mE}{\hbar^2}}; x > 0 \text{ 时, } V=V_0, \psi_2 = Fe^{ik_2x}, k_2 = \sqrt{\frac{2m(E-V_0)}{\hbar^2}}$$

$$\text{又, } \psi_1(0) = \psi_2(0) \Rightarrow A+B=F; \psi_1'(0) = \psi_2'(0) \Rightarrow ik_1A - ik_1B = ik_2F$$

$$\text{整理得 } B = \frac{k_1-k_2}{k_1+k_2}A, R = \frac{|B|^2}{|A|^2} = \left(\frac{k_1-k_2}{k_1+k_2}\right)^2$$

(c) 证 $E > V_0$ 时, $T = \sqrt{\frac{E-V_0}{E}} \cdot \frac{|F|^2}{|A|^2}$, 求 $E < V_0$ 时 T

$$E > V_0 \text{ 时, } j_i = \frac{\hbar k_1}{m} |A|^2, j_t = \frac{\hbar k_2}{m} |F|^2$$

$$T = \frac{j_t}{j_i} = \frac{k_2}{k_1} \cdot \frac{|F|^2}{|A|^2} = \sqrt{\frac{E-V_0}{E}} \cdot \frac{|F|^2}{|A|^2} \text{ 得证}$$

$$E < V_0 \text{ 时, } x > 0, \psi_2 = Ce^{-kx}, \text{ 无透射, } T=0$$

(d) $E > V_0$, 求 T , 证 $T+R=1$

$$E > V_0 \text{ 时, } T = \sqrt{\frac{E-V_0}{E}} \cdot \frac{|F|^2}{|A|^2}, \text{ 已知 } ik_1A = (ik_2 + ik_1)F, \text{ 故 } T = \frac{4k_1k_2}{(k_1+k_2)^2}$$

$$\text{已知 } R = \left(\frac{k_1-k_2}{k_1+k_2}\right)^2, T+R=1$$